Camille Billouard

PhD in AI (Dec. 2026) | 3D Vision & Neural Scene Representations

Paris, France | Personal website | GitHub | Google Scholar

Summary

AI researcher working on neural 3D representations of the physical world: reconstruction with NeRF and Gaussian Splatting from real imagery, multi-view geometry and camera models, scaled to kilometre-scale scenes. I design GPU-efficient, memory-bounded training pipelines (PyTorch, custom CUDA, HPC), evaluate them rigorously against ground truth, and release reproducible open-source code.

Experience

PhD Researcher, *CNES (French Space Agency) & LASTIG, IGN (French Mapping Agency)* — Oct 2023 – Oct 2026 (3 years)
Paris, France

- Formalised compute-independent scalability for neural scene representations (linear time, bounded memory, preserved quality, no reliance on larger hardware) and derived a **fully scalable pipeline** on real multi-view satellite data.
- Built **Snake-NeRF**: out-of-core tiling that reconstructs **52 km²** of Pléiades Neo imagery (442 tiles) **on a single commercial 16 GB GPU**, with bounded peak memory (within 1% of the analytical prediction), 5× lower host memory and near-linear runtime.
- Built **SAT-NGP**: transient-free multi-date NeRF training **86–200× faster** (10–20 h down to 4–14 min) with elevation error within 0.19 m of the state of the art; open-source code reused by other groups as a reference baseline.
- Wrote custom CUDA / C++ kernels integrated into PyTorch for the full rendering pipeline (ray generation from RPC camera models, segmented volume rendering, ray sorting).
- Designed the 3D evaluation protocol against LiDAR HD ground truth: metric choice (elevation error, completeness, masking of vegetation and changing areas), comparison pipeline (co-registration with xDEM, resampling) and test-set definition.

Earlier experience

- **Research Intern**, CNES (French Space Agency), Toulouse (2023, 6 months): NeRF-based 3D reconstruction from multi-date satellite images.
- **Research Intern**, IRIT (Toulouse Institute of Computer Science Research) (2022, 4 months): fast 3D super-resolution for dental CBCT.
- **Software Engineering Intern**, CGI, Bordeaux (2021, 6 months): Java / Spring Batch jobs for banking process automation.

Publications & Talks

C. Billouard, D. Derksen, A. Constantin, B. Vallet. *Tile and Slide: A New Framework for Scaling NeRF from Local to Global 3D Earth Observation.* ICCV Workshops, Hawaii, 2025. arXiv:2507.01631

C. Billouard, D. Derksen, E. Sarrazin, B. Vallet. *SAT-NGP: Unleashing Neural Graphics Primitives for Fast Relightable Transient-Free 3D Reconstruction from Satellite Imagery.* IGARSS, Athens, 2024. doi

K. Specogna, **C. Billouard**, et al. *Applications des champs de radiance neuronaux au mode vidéo de CO3D.* Revue Française de Photogrammétrie et de Télédétection (RFPT), 2026.

Talks: ICCV Workshops (Hawaii, 2025), IGARSS (Athens, 2024), ESA Living Planet Symposium, ORASIS, SFPT, Meet'Up ML (Nantes).

Supervision

Co-supervised two final-year MSc research interns: **3D reconstruction from CO3D satellite video** (K. Specogna, published in RFPT 2026) and **NeRF-based semantic segmentation** (C. Herbin).

Education

PhD in Artificial Intelligence — 2023 – Dec. 2026
Université Gustave Eiffel, CNES, LASTIG, IGN — *Paris, France*

MSc in Artificial Intelligence: Foundations and Applications (IAFA) — 2021 – 2023
Université de Toulouse, specialisation in Computer Vision & 3D — *Toulouse, France*

BSc in Computer Science, research track — 2021
Université de Lille — *Lille, France*

BSc in Mathematics & Computer Science — 2020
Université de Bordeaux — *Bordeaux, France*

Technical Skills

3D Vision: 3D/4D reconstruction, multi-view geometry, camera pose refinement (RPC bundle adjustment), camera models (pinhole, RPC), NeRF (nerfstudio), Gaussian Splatting (LichtFeld Studio), DSM evaluation (xDEM, LiDAR HD, Pléiades Neo)

ML & Systems: PyTorch, custom CUDA kernels, GPU memory profiling, training on SLURM / HPC clusters

Programming: Python, C++, CUDA, SQL, Java, Bash

Languages & interests: French (native), English (fluent); climbing, scuba diving, astrophotography